# SegCLIP: Multimodal Visual-Language and Prompt Learning for High-Resolution Remote Sensing Semantic Segmentation

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***Abstract*— Remote sensing semantic segmentation is considered a key step in the intelligent interpretation of high-resolution remote sensing (HRRS) images, with widespread applications in fields such as hazard assessment, environmental monitoring, and urban planning. Recently, numerous deep learning-based semantic segmentation methods have emerged, achieving significant breakthroughs. However, the majority of current research still concentrates on representation learning in the visual feature space, with the potential of multimodal data sources yet to be fully explored. In recent years, the foundational visual language model, namely contrastive language–image pretraining (CLIP), has established a new paradigm in the visual field, demonstrating excellent generalization capabilities and deep semantic understanding across a variety of tasks. Inspired by prompt learning, we propose a prompting approach based on linguistic descriptions to enable CLIP to generate semantically distinct contextual information for remote sensing images. We introduce the SegCLIP network architecture, a novel framework specifically designed for semantic segmentation of HRRS images. Specifically, we have adapted CLIP to extract text information, thereby guiding the visual model in distinguishing among classes. Additionally, we have designed a cross-modal feature fusion (CFF) module that integrates linguistic and visual semantic features, ensuring semantic consistency across modalities. Finally, we have fully exploited the potential of text data and have used additional real text to refine ambiguous query features. Experimental evaluations confirm that the method exhibits superior performance on the LoveDA, iSAID, and UAVid public semantic segmentation datasets.**

***Index Terms*— Attention mechanism, contrastive language–image pretraining (CLIP), prompt learning, remote sensing, semantic segmentation.**

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## I. Introduction

SEMANTIC segmentation of high-resolution remote sensing (HRRS) images, a critical component of modern space information technology, is increasingly essential in various fields, including global change monitoring [1], resource management [2], environmental protection [3], urban planning [4], and disaster emergency response [5]. Remote sensing semantic segmentation, as a fundamental computer vision task, seeks to precisely classify each pixel of remote sensing images into classes with specific semantic information. Historically, semantic segmentation relied on techniques such as threshold segmentation, edge detection, or traditional machine learning [6], [7], [8]. These methods often struggled with complex scenes and dynamic environments, and they lacked the capability to enhance segmentation quality and automation due to limited manual feature expression, poor generalization, and low accuracy. With advancements in artificial intelligence, deep learning-based HRRS semantic segmentation has garnered significant interest from researchers.

With the development of deep learning [9], [10], significant advancements have been made in image semantic segmentation. The full convolutional network (FCN) [11], as the first model to enable end-to-end learning in semantic segmentation, leverages skip connections to fuse deep and shallow information, markedly enhancing segmentation accuracy. U-Net [12], with its symmetrical encoder–decoder structure, effectively integrates image context and local features through skip connections, thereby further enhancing segmentation accuracy. DeepLabv3+ [13] employs the atrous spatial pyramid pooling module to effectively blend multiscale features, significantly boosting semantic segmentation performance. Cross fusion net (CF-Net) [14] introduces a cross-fusion module for HRRS images, expanding the receptive field of low-level features and effectively enhancing small target segmentation. A relation-augmented fully convolutional network (S-RA-FCN) [15] enhances relational feature representations by analyzing global relationships between spatial locations and feature maps, improving FCN architecture performance. To capture richer contextual information, transformers [16], [17] and their optimized variants [18], [19], [20], [21] have been developed. Transformer-based architectures [22], [23], [24], [25] achieve excellent results in HRRS image semantic segmentation. The

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remote sensing segmentation framework (RSSFormer) [26] introduces a hybrid attention module that combines multihead self-attention with the convolutional block attention module (CBAM) [27], adaptively enhancing foreground saliency features for improved segmentation accuracy. CNN and multiscale transformer fusion network (CMTFNet) [28] constructs a decoder for multiscale, multihead self-attention modules to effectively enhance the multiscale global context information and channel information. Although these methods excel in feature discriminative ability, focusing on single-modal data neglects the rich semantic potential of multimodal data, leading to a performance bottleneck. Consequently, developing a multimodal HRRS semantic segmentation framework is essential to overcome this limitation.

With the increasing diversity of sensors and the expanding volume of remote sensing images, the application of multimodal approaches to semantic segmentation tasks has garnered growing attention [29]. Common remote sensing data include multispectral images, laser radar (LiDAR), and synthetic aperture radar (SAR) data. To fully exploit the distinct characteristics of these datasets, researchers have developed effective fusion strategies. Pan et al. [30] demonstrated that incorporating LiDAR data into 2-D multispectral images effectively addresses variations in building height. Li et al. [31] developed a fusion architecture for land cover classification, employing neural architecture search (NAS) to fuse hierarchical optical features with SAR data. Kang et al. [32] enhanced SAR image segmentation performance through knowledge transfer from optical images.

Recently, visual linguistic representation learning has emerged as a significant area of interest in computer vision research, aiming to extract unified cross-modal feature representations from image–text pairs using deep learning techniques. This approach enhances various multimodal tasks. For instance, Li et al. [33] aligned textual elements with the corresponding image regions, thereby enhancing the association between linguistic and visual components. In remote sensing, Rahhal et al. [34] introduced a multilingual transformer model that enhances cross-modal retrieval accuracy by integrating multilingual features and improving understanding capabilities. Particularly, contrastive language–image pretraining (CLIP) [35] has demonstrated notable success in visual classification tasks, achieving high accuracy with minimal or no annotations, guided by linguistic supervision. Dong et al. [36] leveraged CLIP to notably enhance the accuracy and efficiency of change detection in remote sensing images by integrating it with descriptive text information. Rao et al. [37] and Yang et al. [38] significantly enhanced segmentation performance by incorporating CLIP into semantic segmentation networks through pixel–text alignment. However, these methods typically rely on low-resolution image features aligned with the corresponding text embeddings. As low resolution leads to loss of detail and boundary blurring, these methods are suboptimal.

To address the aforementioned issues, we integrate CLIP into mask classification and propose a novel multimodal HRRS semantic segmentation framework, SegCLIP. Unlike the pixel–text matching method, we propose a novel approach that uses text information as a query condition. In this approach, each mask determines whether the corresponding region belongs to the object described by the query text. These masks not only accurately identify regions of a particular class but also preserve crucial boundary details, which is particularly important for distinguishing adjacent classes. Specifically, inspired by prompt learning, we use CLIP to generate linguistic prompts that describe remote sensing images. The model undergoes extensive training to develop sophisticated cross-modal mapping capabilities, enabling it to effectively use text information related to image content as prior knowledge. To further enhance multimodal fusion in semantic segmentation of remote sensing images and ensure semantic consistency between visual and linguistic features, we designed a cross-modal feature fusion (CFF) module. This module coordinates and integrates high-level semantic features from remote sensing images with their corresponding text descriptions, which effectively solves information heterogeneity between modalities and enhances both information complementarity and synergistic understanding across modalities. On this basis, in order to make full use of the advantages of text information, these real text prompts are cleverly introduced to effectively guide and optimize the feature expressions that are difficult to distinguish or ambiguous in meaning. The contributions of this article are as follows.

1) In this article, we integrate a multimodal visual-language framework into a semantic segmentation network based on mask classification, which employs masks to preserve both detailed and boundary information of categories.
2) To address the critical challenge of cross-modal feature alignment, we have developed a specialized CFF module for multimodal remote sensing semantic segmentation tasks. This module ensures semantic consistency between visual features extracted from remote sensing images and corresponding linguistic textual features.
3) In order to fully explore the potential value of text information, we propose an additional text information description to guide ambiguous query features, which only increases the amount of computation during training and does not introduce additional computation during inference.
4) We conducted extensive experimental evaluations on three public HRRS semantic segmentation datasets: LoveDA, iSAID, and UAVid. These datasets vary in size, class, and mode and aim to validate the effectiveness of the proposed SegCLIP model.

## II. Related Work

### A. Deep Learning-Based Semantic Segmentation

The goal of semantic segmentation is to assign a semantic label to each pixel in a given image so that each pixel is correctly labeled with its corresponding class. Unlike image categorization, the semantic segmentation task faces difficulties due to intraclass and contextual variations [39]. Current deep learning-based semantic segmentation methods can be broadly categorized into two main groups: segmentation methods based on per-pixel classification and mask-based

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classification. The core concept of the former originates from the FCN proposed by Long et al. [11], a work that applied deep learning to image pixel-level semantic annotation for the first time and realized end-to-end learning from image input to pixel-level output. Based on this, a series of significant improvements and optimizations have emerged [12], [13], [40], [41]. Unlike per-pixel classification, where mask classification models predict class-specific masks for each object instance in the image instead of classifying each pixel independently, mask classification is widely used in instance-level segmentation tasks. Mask region-based convolutional neural network (Mask R-CNN) [42] combined an integrated approach of object detection and pixel-level segmentation. It introduced a branching structure for generating fine pixel-level masks based on the fast region-based convolutional neural network (Faster R-CNN) [43], making it suitable not only for semantic segmentation but also for instance-level segmentation tasks. Detection transformer (DETR) [44] introduced an ensemble prediction mechanism that can generate class prediction results for each region with the help of object queries, thus realizing an end-to-end semantic segmentation method with mask probability. Mask classification transformer (MaskFormer) [45] and masked-attention mask transformer (Mask2Former) [46] unified semantic segmentation and instance segmentation, transforming the per-pixel classification problem into a mask-predictive one for problem processing. Yu et al. [47] redefined the relationship between pixels and object queries. They considered object queries as clustering centers with learnable embedded vectors and proposed redefining cross-attention learning as a clustering process. In remote sensing, the mixed-mask transformer (MMT) [48] introduced mask classification into remote sensing image decoding for the first time. It proposed a transformer structure that fused multiple mask information, capable of capturing and aggregating multilevel and multitype features in complex remote sensing scenes. The multiscale prototype contrast network (MPCNet) [49] proposed a mask classification network based on comparing prototypes and introduced a new multiscale prototype transformer decoder. This network effectively integrated information from feature maps and learnable class labels, from which prototypes reflecting the specificity of dynamic scenes were extracted and used as classifiers.

Although the above methods have achieved remarkable results, they rely only on single-modal data, which undoubtedly limits their performance potential. Therefore, we present a method for semantic segmentation of HRRS images that deeply explores and fully uses multimodal fusion strategies based on mask classification techniques.

### *B. Multimodal Prompt Learning*

Over the past decade, the learning paradigm for computer vision tasks has been predominantly characterized by the "pretraining and fine-tuning" approach. Specifically, models are pretrained on natural images either by supervised learning [16], [50], [51] or self-supervised learning [52], [53], [54]. Subsequently, on the basis of the pretrained model, the model parameters are further fine-tuned and optimized in a targeted manner for various specific downstream tasks, serving as a way to adapt to different visual understanding and recognition needs. For a long time in the past, this process has mainly relied on pretraining with large natural image datasets like ImageNet [55]. Over time, the learning paradigm has shifted away from the traditional "pretraining + fine-tuning" approach to a new phase: "pretraining and prompting" [56], [57], [58]. In this new paradigm, researchers have shifted their focus to reformulating downstream tasks using prompts to redesign the inputs to be consistent with the original pretrained task [59], [60], effectively minimizing the semantic bias between the pretrained model and the task-specific fine-tuning.

In recent years, an increasing number of researchers have used multimodal learning for computer vision tasks [61], [62], [63]. Lu et al. [64] employed a coattentive mechanism to facilitate interaction between the dual streams, allowing the model to take into account the correlation between the two modalities during processing and thus learn a joint visual–verbal representation. Additionally, Tan and Bansal [65] jointly modeled the information from both modalities through a shared transformer structure, learning a unified and interrelated cross-modal representation. Li et al. [66] proposed a method for combining the learning of visual and linguistic representations, prealigning them through momentum distillation techniques to improve the fusion of information from both modalities. Huo et al. [67] introduced a large-scale multimodal pretraining method aimed at effectively constructing and bridging the link between vision and language through joint training of image and text data. Among these, CLIP [35], a landmark in self-supervised and multimodal deep learning, uses a large-scale comparative learning mechanism. This approach trains a model to effectively pair images with related text descriptions, spanning from everyday natural scenes to specialized areas like remote sensing and medical imaging.

Influenced by CLIP, Rao et al. [37] classified each pixel in an image using the CLIP model to achieve more precise and dense semantic segmentation predictions by converting image–text matching into pixel–text matching. As the above methods generated only single-scale feature maps and did not provide multiscale maps, Yang et al. [38] built upon this foundation and achieved semantic segmentation of remote sensing images by integrating multimodal and multiscale information. Nonetheless, these methods continue to encounter several challenges. Because pixel–text matching primarily depends on low-resolution image features, it can result in unclear boundaries between categories. At low resolutions, subtle texture and boundary information are easily lost, complicating the differentiation between pixel classes within the same region. In contrast to other approaches, our SegCLIP integrates mask classification and text-based vision for remote sensing semantic segmentation. Compared to pixel–text matching, each mask not only accurately identifies the location of the target object but also better preserves its boundary information.

## III. METHOD

In this section, we initially elaborate on the SegCLIP framework, a learning approach for the HRRS semantic segmentation task that uses CLIP for semantic prompting. In Section III-A, we provide an overview of the overall

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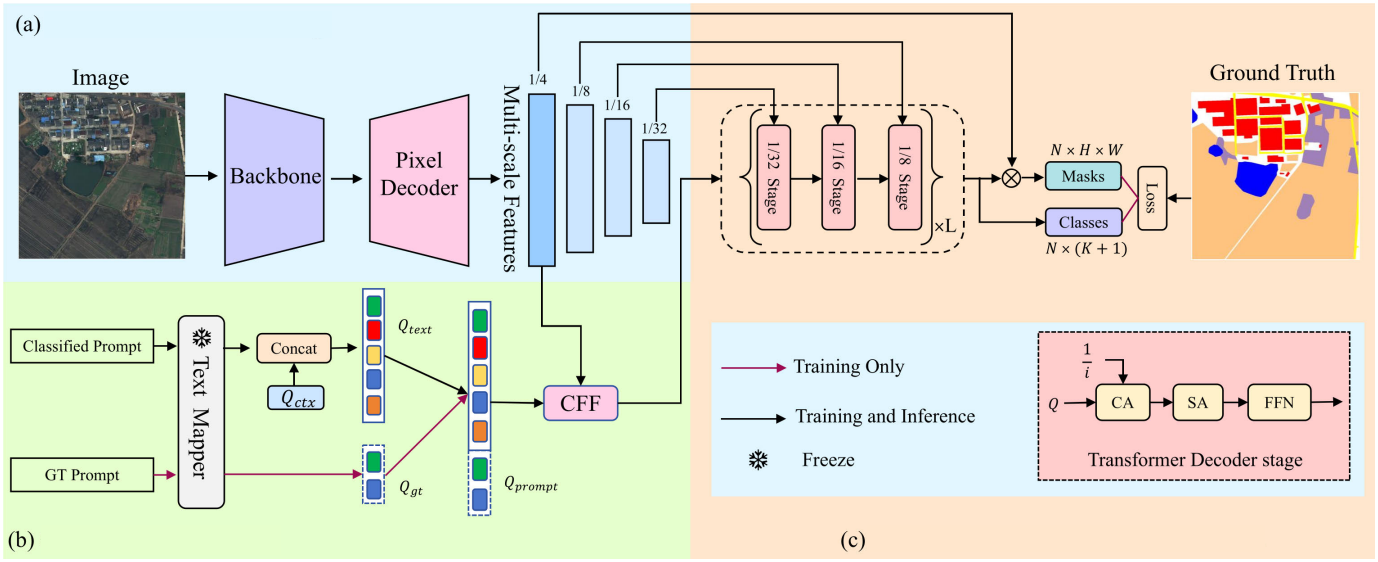



Fig. 1. SegCLIP overall framework: the framework is divided into three main components: (a) extracting multiscale features; (b) generating linguistic descriptions for subsequent text prompts; and (c) generating categorizations and their corresponding masks. The GT prompt, derived from GT, refines the query scope further. To ensure model robustness, the GT prompt is only used during training and is removed during inference, as indicated by the red arrow.

architecture of SegCLIP, which is divided into three core components: multiscale feature modeling, cross-modal text prompt and transformer decoder, and class prediction. Our work focuses on cross-modal text prompt and transformer decoder, namely text mapping prompt (TMP), CFF module, and ground-truth text guiding (GTG). These three components are described in detail in Sections III-B–III-D, respectively.

### A. Overall Structure

The overall architecture of SegCLIP is shown in Fig. 1, comprising three main parts: multiscale feature modeling, cross-modal text prompt, and transformer decoder. Initially, in the multiscale feature modeling stage, ResNet-50 [50] is selected as the backbone network structure to extract the multilevel feature representation of the image. Simultaneously, the pixel decoder is constructed utilizing the deformable self-attention mechanism. When the remote sensing image is processed by ResNet-50, four feature maps of different levels are generated with downsampling ratios of 1/4, 1/8, 1/16, and 1/32, which are then refined by the pixel decoder to achieve accurate prediction at each pixel level.

Cross-modal text prompt mainly includes TMP, CFF module, and GTG, which is also the main part of our work. Although single-modal semantic segmentation methods for remote sensing images have achieved remarkable results, performance enhancement faces a bottleneck due to inherent limitations. Therefore, we utilize the unsupervised classification capability of the CLIP model to generate text prompts and use them for multimodal input data, focusing on semantic segmentation. We only use the text encoder of CLIP and freeze its parameters, and this process is done by TMP. The representations and structures of text and image data differ significantly, and directly using text features to guide semantic feature extraction in images may lead to difficulties in model learning. Therefore, we combine text and visual features through the CFF module to ensure information consistency. In addition, within the cross-modal text prompt, we introduce ground-truth (GT) text information through GTG to improve the accuracy of the target query.

Then, after multiple layers of transformer decoder operations, the class prediction vector ($P_{\text{class}}$) with a dimension of $N \times (K+1)$ is finally obtained, where $N$ represents the number of queries, $K$ represents the total number of classes, and $K+1$ is due to the inclusion of the "background" class representing no specific target. In addition, the binary mask prediction $P_{\text{mask}}$ has dimensions $N \times H \times W$, which is obtained by performing an element-by-element product operation on the mask embedding and the class prediction vector $P_{\text{class}}$, where $H$ and $W$ represent the height and width of the image, respectively. Finally, the final semantic segmentation result is obtained by combining $P_{\text{class}}$ and $P_{\text{mask}}$ through matrix operations.

To better understand the workflow of SegCLIP, we describe the method in detail in Algorithm 1.

### B. Text Mapping Prompt

To address the absence of specific linguistic descriptions for each image in remote sensing semantic segmentation datasets, we construct an adaptive method for generating such descriptions, aiming to harmonize mask classification with visual text. Through the introduction of a joint visual–linguistic learning mechanism, we transform cross-attention learning into a process of exploring the intrinsic connection between visual data and linguistic descriptions. Compared to the existing CLIP variants [68], [69], our research emphasizes applications in closed-set recognition scenarios. Specifically, we choose the LoveDA dataset [70] as an example, which covers seven core classes: background, building, road, water, barren, forest, and agriculture. For the $n$ classes, we construct the corresponding text descriptions, as shown in Fig. 2. The template "a photo with a class" is used for each class name as an indicative prompt for the foreground content of remote sensing images.

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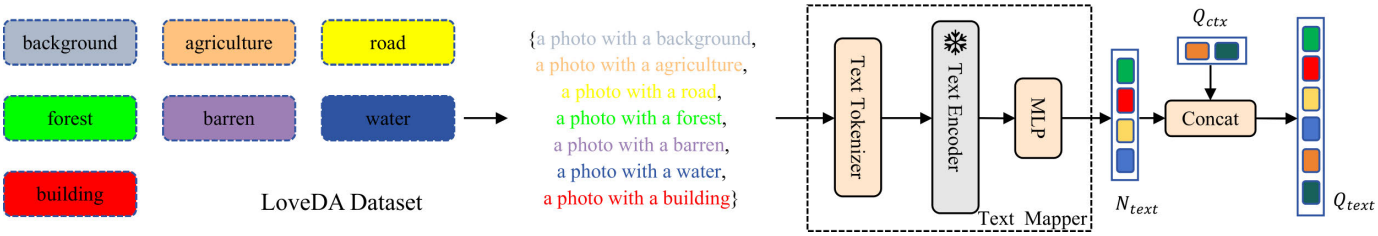



Fig. 2. Using LoveDA as an example, LoveDA comprises seven classes: background, agriculture, road, forest, barren, water, and building. We use the template "a photo with a {class}" to compile a list of text descriptions for each class, where "class" denotes the class name. The text mapper includes a text tokenizer, a frozen text encoder, and a three-layer multilayer perceptron (MLP). This component generates visually pertinent text prompt query embeddings ($N_{\text{text}}$), which is subsequently integrated with learnable textual context embeddings ($Q_{\text{ctx}}$) to form text prompts ($Q_{\text{text}}$).

**Algorithm 1** Workflow of the Proposed SegCLIP

**Input:** The input image $\boldsymbol{x} \in \mathbb{R}^{3\times H\times W}$ and corresponding label $\boldsymbol{y} \in \mathbb{R}^{H\times W}$.
**Output:** The semantic segmentation results $\boldsymbol{o} \in \mathbb{R}^{H\times W}$.
**CFF:** Cross-modal feature fusion module.

**repeat:**
  **Multi-scale Feature Modeling:**
  Generate basic feature map $\{S_i\}_4^{i=1}$ from the backbone.
  Obtain fused multi-scale feature map $\{F_i\}_4^{i=1}$ from the pixel decoder.
  **Cross-modal Text Prompt:**
  **Branch 1:** The input is the class names of a dataset.
  Obtain prompt query embeddings $N_{text}$ for $n$ classes from the text mapper.
  Connect learnable textual context embeddings $Q_{ctx}$ and prompt query embeddings $N_{text}$ to obtain textual query terms $Q_{text}$.
  **Branch 2:** The input is class names that exist for the label $y$ and is used only for training.
  Repeat step 7 to obtain the label text information $Q_{gt}$.
  **Then:**
  Use CFF to obtain text prompt queries $Q_{prompt}$. If training, $Q_{prompt}$ = CFF(Concat($Q_{text}$, $Q_{gt}$)); otherwise, $Q_{prompt}$ = CFF($Q_{text}$).
  **Transformer Decoder:**
  Update the queries $Q_{prompt}$ using the multi-scale feature map $\{F_i\}_3^{i=1}$.
  Generate the class prediction and mask prediction using the last scale $F_4$ with queries $Q_{prompt}$.
  **Loss:**
  Calculate overall loss $L$ following Equation 19.
**until** convergence
The final output segmentation map ($\boldsymbol{o}$) is generated by matrix multiplication of the class prediction and the mask prediction.

The text is structured as text = [$a$ photo with $a$ {class$_i$}], where $i \in [1, 7]$. Subsequently, "a photo of" is used to expand to the specified dimension. The text description of each sample varies across datasets. Subsequently, these natural language descriptions are transformed into text-encoding features using a sequence data processing method known as a text tokenizer. These features are then mapped into text embeddings, which contain contextual meanings, using a text encoder (encoder$_{\text{text}}$). The encoder is an architecture that contains 12 layers of transformers. As a result, the text embeddings contain not only the binary mask information in the image but also the probability distribution of each class, thus providing rich semantic information for the model. We define the text tokenizer as $T_t$, and we define the prompt query embeddings ($N_{\text{text}}$) of class $n$ by $T_t$ and encoder$_{\text{text}}$. The whole process can be formulated as follows:

$$N_t = T_t(\text{text}) \tag{1}$$

$$N_{\text{text}} = \text{encoder}_{\text{text}}(N_t). \tag{2}$$

To further enhance the semantic understanding capability of the model, we introduce a set of learnable textual context embeddings $Q_{\text{ctx}}$ and combine them with the text embeddings generated by the encoder, which in turn generates $N$ textual query terms ($Q_{\text{text}}$). The motivation is to extract a unified context of text within sample images [71]. $Q_{\text{ctx}}$ interacts with multiscale features of the input during forward propagation and progressively optimizes the context through multiple iterations (i.e., multiple decoder layers). Self-attention within the decoder further enables $Q_{\text{ctx}}$ to thoroughly explore the word embedding space, thereby enhancing the learning of task-relevant context. The computational formula is as follows:

$$Q_{\text{text}} = \text{Concat}(N_{\text{text}}, Q_{\text{ctx}}) \tag{3}$$

where Concat denotes the splicing operation.

### C. CFF Module

Due to the inherent complexity of remote sensing images, the text information extracted by the text encoder may struggle to correspond accurately with specific remote sensing images, posing a challenge for multimodal remote sensing semantic segmentation. To address this problem, the interaction between visual features and linguistic text information becomes particularly important. To achieve this goal, we design a CFF module aimed at efficiently integrating high-level semantic features of images with text embedding features to enhance the synergy and relevance between different modalities. It is worth noting that typically, the number of classes ($n$) is smaller than the number of queries ($N$). Consequently, when constructing text prompts, the original text embeddings may contain redundant "blank" information, which is unhelpful for guiding prompts in segmentation tasks. Specifically, as shown

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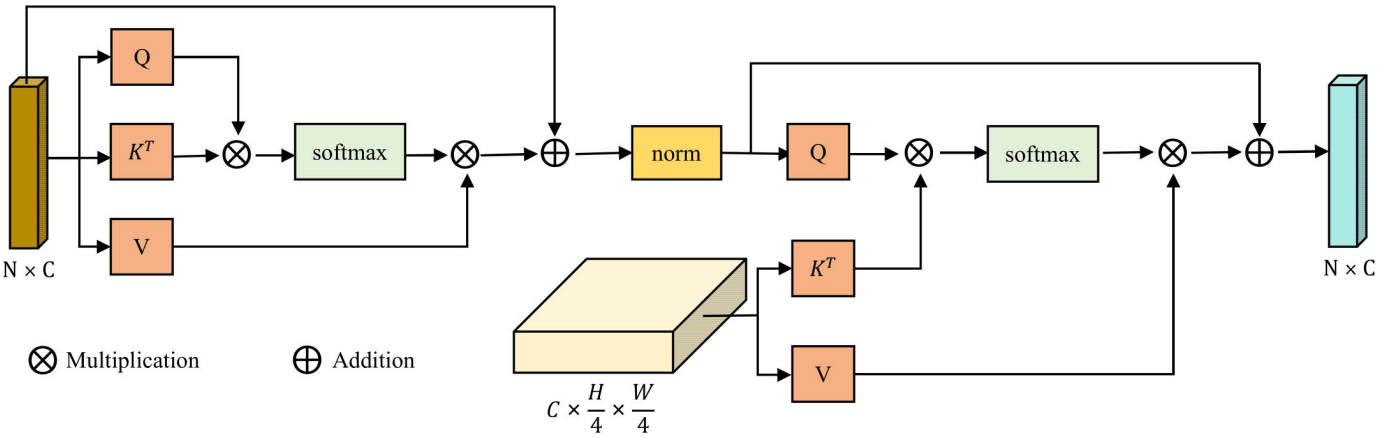



Fig. 3. Schematic of the CFF module. The inputs to the CFF module include query terms processed by the text mapper and visual features refined through multiscale feature modeling. $N \times C$ denotes $N$ queries with dimension $C$.

in Fig. 3, we employ the attention mechanism to facilitate the interaction and integration of information by each location feature according to its sequence, aiming to enhance and harmonize the consistency of internal feature representations, thereby suppressing the expression of redundant information. This approach is particularly advantageous for information that relies heavily on textual context. The computational formula is as follows:

$$Q = \text{linear}(Q_{\text{text}}) \tag{4}$$

$$K = \text{linear}(Q_{\text{text}}) \tag{5}$$

$$V = \text{linear}(Q_{\text{text}}) \tag{6}$$

$$Q_{\text{text}_1} = Q_{\text{text}} + \text{Softmax}\left(\left(QK^T\right)/\sqrt{d}\right) \cdot V \tag{7}$$

$$Q'_{\text{text}_1} = \text{norm}\left(Q_{\text{text}_1}\right) \tag{8}$$

where $Q_{\text{text}}$ represents the input query terms. $Q$, $K$, and $V$ are obtained from $Q_{\text{text}}$ through linear mapping. Norm indicates the normalization method, aimed at enhancing the stability of the algorithm.

Language information offers semantic descriptions and scene interpretations of remote sensing images, while visual features provide details regarding specific characteristics and spatial relationships. These two types of information complement each other and enhance the analytical capabilities of the model. To integrate the intrinsic associations and ensure semantic consistency across different modalities, we use the cross-attention mechanism to align the high-level semantic features ($\Gamma \in \mathbb{R}^{C\times(H/4)\times(W/4)}$) of images with the text embedding features ($Q'_{\text{text}_1}$), with the aim of enhancing the close correlation of multimodal data. Notably, the advanced semantic features of the image use the 1/4 feature scale. This is because the 1/4 feature scale contains multiscale information after refinement by the pixel decoder and preserves detailed features better than other scales. The calculation formula is as follows:

$$Q = \text{linear}\left(Q'_{text_1}\right) \tag{9}$$

$$K = \text{linear}(\Gamma) \tag{10}$$

$$V = \text{linear}(\Gamma) \tag{11}$$

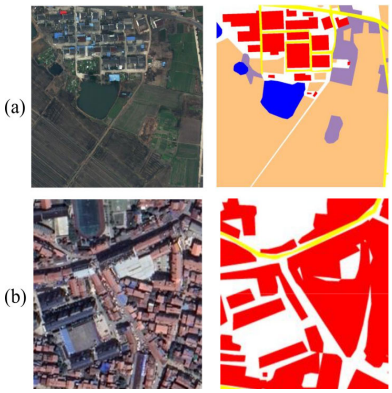



Fig. 4. We selected two images to illustrate the limitations of TMP. The LoveDA dataset includes the following classes: "background," "agriculture," "road," "forest," "barren," "water," and "building." Image (a) includes "background," "agriculture," "road," "barren," "water," and "building." Image (b) includes only "background," "road," and "building."

$$Q_{\text{text}_2} = Q'_{\text{text}_1} + \text{Softmax}\left(\left(QK^T\right)/\sqrt{d}\right) \cdot V \tag{12}$$

$$Q'_{\text{text}_2} = \text{norm}\left(Q_{\text{text}_2}\right) \tag{13}$$

where $Q$ represents the text features obtained by the linear mapping of $Q'_{\text{text}_1}$, which contains rich textual semantic information, $K$ and $V$ represent the visual semantic features obtained by multiscale feature modeling to obtain the 1/4 feature scale $\Gamma$, and then apply a linear mapping. These features encompass abundant contextual information and have the ability to capture and express intricate details and structural relationships within the image. Aligning these features through the cross-attention mechanism enables the model to better understand the inherent relationship between text and image.

### D. Ground-Truth Text Guiding

In Section III-B, we use class labels from the closed set as text prompts to precisely define the query scope. However, this process fails to fully exploit the potential of text information. Specifically, although we specify the text prompts of "background," "agriculture," "road," "forest," "barren," "water," and "building" for the remote sensing image in Fig. 4, corresponding to labels 1–7, only the labels "background," "road," and

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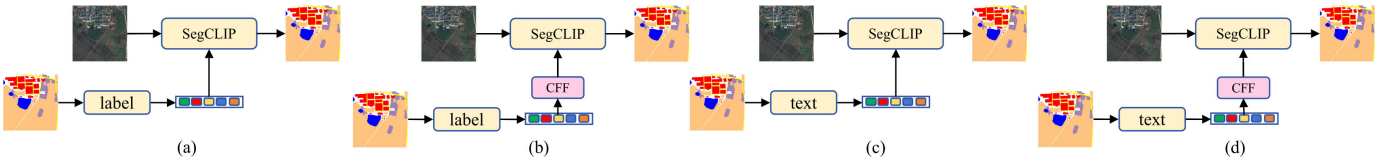



Fig. 5. We offer four solutions. (a) and (b) Use labels extracted from GT, indicating the current class associated with the GT. (c) and (d) Employ GT to extract linguistic descriptions for the class names in the labels.

"building" are relevant in (b), and the rest of the prompts are irrelevant, resulting in a low-prompt utilization rate. Although the current method defines the query scope through labels, it cannot efficiently use text prompts. How can one formulate more accurate guidance? We can naturally consider using GT as a query guide, that is, employing GT as additional queries. To ensure the robustness of the model, we add additional real text information only during training, and this information does not participate in any calculations during inference. This approach ensures that the rich semantic information of text prompts can be effectively learned during model training, thus enhancing the generalization ability of the final model in practical applications. In this context, we present four methods for reference. As shown in Fig. 5, the prompt content for (a) and (b) consists of the class (typically an integer from 1 to $n$), which serves as additional queries. (c) and (d) employ GT to acquire language descriptions as prompt content, utilizing the respective prompt content as additional queries. Experimental results demonstrate that employing method (d) yields superior performance. Therefore, we employ (d) as additional queries in the model. This method aims to enable the model to effectively leverage this authentic information, thereby facilitating clearer guidance for the model from the outset of the learning process. Specifically, initially, we use GT to extract the foreground portion of the remote sensing image and acquire the GT text information describing the foreground ($\text{text}_{\text{gt}}$). For ease of calculation, assume that GT contains the foreground of the first $t$ classes. Consequently, $\text{text}_{\text{gt}}$ is formulated as [$a$ photo with $a$ {class$_i$}], where $i \in [1, t]$. Subsequently, $\text{text}_{\text{gt}}$ undergoes processing by the text tokenizer ($T_t$) and the text encoder ($\text{encoder}_{\text{text}}$) to generate the text mapping ($Q_{\text{gt}}$), which is then embedded as additional queries. Throughout this process, the weight is shared with the original query terms ($Q_{\text{text}}$) to ensure consistency. Subsequently, $Q_{\text{gt}}$ is concatenated with $Q_{\text{text}}$ to create a set of prompt queries ($Q_{\text{cat}}$). Here, $Q_{\text{text}}$ can be represented as $(Q_1, Q_2, \ldots, Q_n)$, and $Q_{\text{gt}}$ can be represented as $(Q_1, Q_2, \ldots, Q_t)$. The calculation steps proceed as follows:

$$Q_{\text{gt}} = \text{encoder}_{\text{text}}\left(T_t\left(\text{text}_{\text{gt}}\right)\right) \tag{14}$$

$$Q_{\text{cat}} = \text{Concat}\left(Q_{\text{text}}, Q_{\text{gt}}\right) \tag{15}$$

where $n$ is the total number of classes and $t$ represents the number of classes the current image has. $T_t$ represents the text tokenizer converter, $\text{encoder}_{\text{text}}$ represents the text encoder, and this part of the process is the same as TMP. Concat represents the concatenation operation. During training, $Q_{\text{cat}}$ is composed of $Q_{\text{gt}}$ and $Q_{\text{text}}$. In the inference stage, we remove $Q_{\text{gt}}$; thus, $Q_{\text{cat}} = Q_{\text{text}}$. Subsequently, the CFF module is used to interactively fuse $Q_{\text{cat}}$ with visual features, thereby unifying

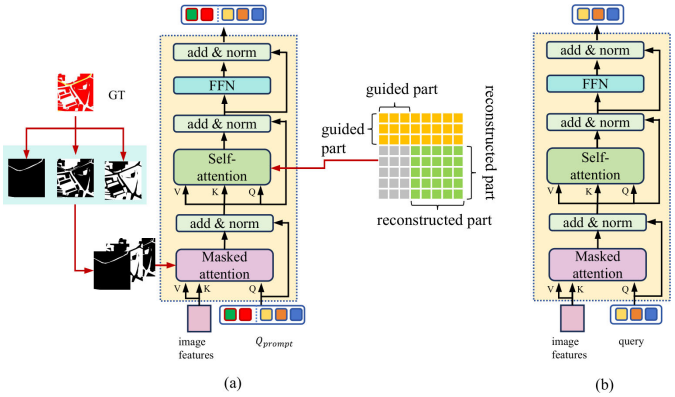



Fig. 6. (a) Structure of the transformer decoder stage during training. The yellow and green grids in the attention mask represent unmasked, and the gray grids represent masked. (b) Structure of the transformer decoder stage during inference. The input contains only the reconstructed part.

the multimodal contextual relationship. This can be expressed as follows:

$$Q_{\text{prompt}} = \text{CFF}(Q_{\text{cat}}). \tag{16}$$

Therefore, during the training phase of the decoder, the queries are divided into two parts: the guided part ($Q_{\text{gt}}$) and the reconstructed part ($Q_{\text{text}}$). The guided part comes from GT, while the reconstructed part comes from the category descriptions in the dataset. Subsequently, we obtain the foreground regions corresponding to each category from GT as binary masks, as shown in Fig. 6. In the mask attention stage, we combine these category-specific binary masks with the original masks and use the binary masks from GT to guide the reconstruction process of the original masks. During the self-attention stage, we introduce an attention mask method [72]. This is because the queries themselves carry label information during the training process. The purpose is to prevent the model from receiving GT information directly from the queries, which leads to information leakage. We denote the attention mask as $A = [a_{ij}]_{WW}$, where $W = M + N$. $M$ indicates the number of GT objects in the bootstrap part, and $N$ denotes the number of queries in the reconstruction part. We designate $M$ rows and columns to represent the bootstrap part and $N$ to represent the reconstruction part. The computational formula is as follows:

$$\begin{cases} 1, & \text{if } \ j < M \text{ and} |\dfrac{i}{M}| \neq |\dfrac{j}{M}|; \\ 1, & \text{if } \ j < M \text{ and } i \geq M; \\ 0, & \text{otherwise} \end{cases} \tag{17}$$

where $a_{ij}$ means that the $i$th query cannot see the $j$th query; otherwise, $a_{ij} = 0$. Assuming the number of queries is $N$,

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then the decoder queries can be expressed as $Q_{\text{prompt}} = \{Q_0, Q_1, \ldots, Q_{N-1}\}$, and the transformer decoder output is $O = \{O_0, O_1, \ldots, O_{N-1}\}$. We use $F$ and $A$ to denote the refined image features after the transformer encoder, and then the decoder equation can be expressed as follows:

$$O = D\big(Q_{\text{prompt}}, \text{Mask}, F|A\big) \tag{18}$$

where Mask is the hard constraint mask. $D$ is the Transformer decoder.

### E. Loss Function

The SegCLIP training process involves calculating two primary losses: the predicted classification loss and the predicted mask loss. The predictive classification loss guides the optimization of textual information, and the predictive mask loss ensures consistency between the segmentation masks generated by the model and the target regions. Specifically [46], we use cross-entropy loss ($L_{\text{cls}}$) for the classification loss, as well as binary cross-entropy loss ($L_{\text{bce}}$) and dice loss ($L_{\text{dice}}$) for the mask loss. The total loss of SegCLIP is the weighted sum of the classification loss and the mask loss, which can be expressed as follows:

$$L_{\text{total}} = \lambda_{\text{cls}} L_{\text{cls}} + \lambda_{\text{bce}} L_{\text{bce}} + \lambda_{\text{dice}} L_{\text{dice}} \tag{19}$$

where $\lambda_{\text{cls}}$, $\lambda_{\text{bce}}$, and $\lambda_{\text{dice}}$ are the corresponding weights, which we empirically set $\lambda_{\text{cls}} = 2$, $\lambda_{\text{bce}} = 5$, and $\lambda_{\text{dice}} = 5$. For no-object predictions, i.e., the "background," $\lambda_{\text{cls}}$ is set to 0.1.

## IV. Experiments

### A. Dataset Information

*1) LoveDA Dataset:* The LoveDA [70] dataset, encompassing both urban and rural regions, is compiled from three Chinese cities: Nanjing, Changzhou, and Wuhan. This dataset, sourced from Google Earth, features a ground sampling distance of 0.3 m and comprises 166 768 annotated objects across 5987 nonoverlapping sections, each measuring 1024 × 1024 pixels. It includes seven annotation classes: background, agriculture, road, water, forest, barren, and building. Specifically, for our experiments, we use 2522 images for training and 1669 images for validation/testing.

*2) iSAID Dataset:* The iSAID [73] dataset is a comprehensive, densely annotated aerial segmentation dataset that includes 655 451 object instances across 15 classes, distributed over 2806 high-resolution images. The classes include ship (SH), storage tank (ST), baseball diamond (BD), tennis court (TC), basketball court (BC), ground track field (GTF), bridge (BR), large vehicle (LV), small vehicle (SV), helicopter (HC), swimming pool (SP), roundabout (RA), soccer ball field (SBF), plane (PL), and harbor (HA). Image sizes vary from 800 × 800 to 4000 × 13000 pixels. In our experiments, we partition this data into 1411 training images, 458 validation images, and 937 testing images. Due to the unavailability of the GT for the test set, we use 1411 training set images for training and 458 images for validation. Each image is overlaid and segmented into subimages of 896 × 896 pixels with a step size of 512 × 512.

*3) UAVid Dataset:* UAVid [74] is a high-resolution unmanned aerial vehicle (UAV) semantic segmentation dataset with remarkable features in large-scale mutation, moving target recognition, and temporal fusion preservation. The dataset contains 42 video sequences shot from an oblique perspective, providing high-resolution images at two spatial resolutions (3840 × 2160 and 4096 × 2160). In the dataset, a total of 420 images are densely annotated into eight different classes to support the semantic labeling task. Among them, 200 images are used for training, 70 images are used for verification, and the remaining 150 images are used for testing. In the experiments, each image is processed by padding and cropped to a patch of 1024 × 1024 pixels.

### B. Experimental Settings

*1) Implementation Details:* The proposed model is implemented using PyTorch 1.13.1 and the Detectron2 framework. The experimental environment consists of Ubuntu 22.04, and a GeForce RTX 4090 is employed to train our model. The number of queries is fixed at 100. To facilitate comparisons with other models, we consider two backbone networks: ResNet-50 [50] and Swin-B [19]. For text prompts, a text encoder with a context length of 77 and 12 layers is used. For visual feature extraction, a pixel decoder with six layers and a transformer decoder with nine layers are used. To reduce computational costs, the multiscale feature dimension of the image and the text embedding dimension are mapped to a lower dimension (256) before the decoder. Notably, the text encoder uses the text encoder component of CLIP and remains frozen during training. The initial learning rate is set to 0.0001, and the multistep learning rate scheduler decreases it twice by 0.1 at each step. For data augmentation, large-scale jitter (LSJ) augmentation [46] with random samples ranging from 0.1 to 2.0 is employed.

*2) Evaluation Metrics:* In this study, two commonly used metrics—the mean intersection over union (mIoU) and the mean $F1$ score (mF1)—are employed to evaluate performance. The mIoU metric reflects the overall segmentation precision, while the mF1 is a composite score of precision and recall. The values of mIoU and mF1 are calculated based on the cumulative confusion matrix as follows:

$$\text{mIoU} = \frac{1}{K}\sum_{i=1}^{K}\frac{\text{TP}_i}{\text{TP}_i + \text{FP}_i + \text{FN}_i} \tag{20}$$

$$\text{mF1} = \frac{1}{K}\sum_{i=1}^{K}\frac{2 * \text{TP}_i}{2 * \text{TP}_i + \text{FP}_i + \text{FN}_i} \tag{21}$$

where $K$ represents the number of classes, and $\text{TP}_i$, $\text{FP}_i$, and $\text{FN}_i$ denote the true positives, false positives, and false negatives, respectively, for objects classified as class $i$.

### C. Ablation Study

A series of ablation experiments are conducted to investigate the importance of each component and parameter setting in our proposed method. The experiments use the LoveDA dataset, employing ResNet-50 as the backbone network with a query count of 100, unless otherwise stated.

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TABLE I
ABLATION STUDY OF THE EFFECTIVENESS OF EACH COMPONENT ON THE LOVEDA DATASET

| Method | TMP | $Q_{ctx}$ | CFF | GTG | mF1(%) | mIoU(%) |
|---|---|---|---|---|---|---|
| Baseline | - | - | - | - | 69.23 | 53.77 |
| | ✓ | - | - | - | 69.90 | 54.45 |
| | ✓ | ✓ | - | - | 70.14 | 54.94 |
| | ✓ | ✓ | ✓ | - | 70.73 | 55.40 |
| | ✓ | ✓ | ✓ | ✓ | 71.44 | 56.27 |

TABLE II
ABLATION STUDY OF THE EFFECTIVENESS OF DIFFERENT $Q_{\text{CTX}}$ VALUES ON THE LOVEDA DATASET

| $Q_{ctx}$ | 0 | 8 | 16 | 32 | 64 |
|---|---|---|---|---|---|
| mF1(%) | 69.81 | 70.13 | 71.44 | 70.25 | 70.00 |
| mIoU(%) | 54.77 | 54.93 | 56.27 | 55.00 | 54.60 |

*1) Impacts of Components on SegCLIP:* To assess the effectiveness of each component, Table I shows the ablation experiments. The first row depicts the baseline; a 0.68% improvement in mIoU is observed after the addition of the TMP, indicating that multimodal text prompts aid in the visual segmentation task. The second row demonstrates an additional 0.49% improvement in mIoU after the addition of the learnable textual context embeddings $Q_{\text{ctx}}$ and the TMP, further highlighting the critical role of uniform contextual information in the sample images for enhancing the performance of the model. The 0.46% improvement in mIoU with the addition of TMP, $Q_{\text{ctx}}$, and CFF demonstrates that CFF is effective in enhancing the coherence between different modalities. In addition, we use GT to restrict the query to a finer range, achieving an improvement of 0.87% in mIoU compared to the previous combination. There is also a gradual increase in the mF1 metric. In summary, our approach is progressive and collectively enhances the model performance.

*2) Impacts of Different Textual Contextual Embeddings $Q_{ctx}$ on SegCLIP:* We have integrated learnable textual context embeddings into TMPs to improve model understanding in complex scenarios. To rigorously determine the optimal value of $Q_{\text{ctx}}$, we maintain a constant number of queries. We conduct a series of experiments to explore the effects of varying $Q_{\text{ctx}}$ on text prompt queries. Notably, LoveDA offers seven classes of text prompts during training. The experimental results, as shown in Table II, indicate that the model performs optimally when $Q_{\text{ctx}}$ is set to 16, significantly surpassing performance metrics under alternative configurations. When $Q_{\text{ctx}}$ exceeds 16, model performance declines, despite an increase in learnable context embeddings. The likely cause is that an excessively large $Q_{\text{ctx}}$ leads to overfitting during the learning process, causing the model to overly focus on specific textual context details at the expense of broader general features, thereby impairing its generalization ability.

*3) Impacts of Different GT Prompts on Text Mapping:* We evaluate the effects of four distinct GT guiding methods on text prompts, as shown in Table III. To ensure experimental fairness, we maintain all other parameters constant. Guiding methods are shown in Fig. 5. Unlike (a), a CFF module is integrated into (b) to ensure consistency across different modalities. The results, shown in the second and third rows, indicate that directly applying GT labels as prior knowledge for query guidance significantly diminishes the natural linkage between textual and visual features, resulting in diminished performance. These findings suggest that GT labels may not be an optimal choice for text prompts. Options (c) and (d) differ from (a) and (b) in that GT text, obtained after foreground information extraction, is employed for additional prompts. The results in the third and fourth rows demonstrate the effectiveness of CFF in unifying the intrinsic associations of different modalities. Scheme (d) is confirmed as the best option in the model architecture, as can be seen from the table. Employing GT text as a priori knowledge can significantly improve the performance of the model, with mIoU increasing from 55.4% to 56.27%. This improvement clearly demonstrates that focusing on the intrinsic association between text and visual content significantly enhances the understanding and segmentation accuracy of the model.

TABLE III
ABLATION STUDY OF THE EFFECTIVENESS OF DIFFERENT GT PROMPTS ON THE LOVEDA DATASET

| Method | GT Label | GT Text | CFF | mF1(%) | mIoU(%) |
|---|---|---|---|---|---|
| SegCLIP | - | - | - | 70.73 | 55.40 |
| | ✓ | - | - | 69.50 | 54.27 |
| | ✓ | - | ✓ | 69.45 | 54.11 |
| | - | ✓ | - | 69.55 | 54.16 |
| | - | ✓ | ✓ | 71.44 | 56.27 |

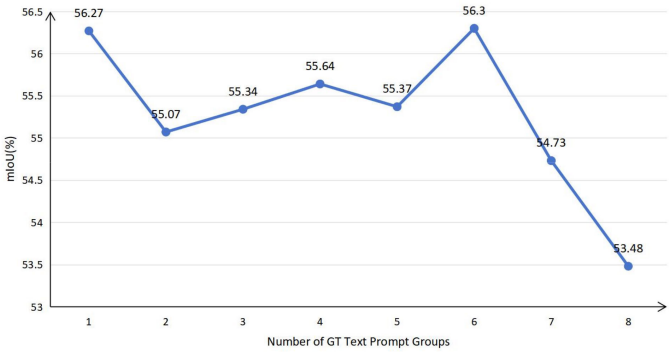



Fig. 7. Analysis of the number of GT text prompt groups. The horizontal axis represents the number of groups, and the vertical axis shows the mIoU percentage.

*4) Impacts of the Number of GT Text Prompt Groups on SegCLIP:* Our study analyzes variations in model performance with differing numbers of GT prompt groups. As shown in Fig. 7, using a single prompt group results in the model achieving an mIoU of 56.27%. Interestingly, increasing the number of prompt groups to two or three leads to a slight decrease in model performance to 55.07% and 55.34%, respectively, indicating that the initial increase in the number of groups does not improve performance. Furthermore, expanding the number of prompt groups to six only marginally increases performance by 0.03% compared to using a single prompt group, a negligible change that introduces unnecessary computational burden.

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TABLE IV
COMPARISON OF COMPLEXITY ON THE LOVEDA DATASET

| Method | Backbone | Params(Mb) | | | FLOPs(Gbps) | | |
|---|---|---|---|---|---|---|---|
| | | Backbone | Text Encoder | Segmentation Head | Backbone | Text Encoder | Segmentation Head |
| Baseline | ResNet-50 | 23.51 | - | 20.50 | 85.88 | - | 174.05 |
| SegCLIP | ResNet-50 | 23.51 | 63.69 | 22.52 | 85.88 | 247.64 | 190.48 |

TABLE V
QUANTITATIVE COMPARISONS WITH THE STATE-OF-THE-ART METHODS ON THE LOVEDA DATASET. THE METRIC FOR ALL CATEGORIES IS IOU

| **Method** | **Backbone** | **Per-class IoU(%)** | | | | | | | **mF1** | **mIoU** |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Background | Building | Road | Water | Barren | Forest | Agriculture | (%) | (%) |
| FCN [11] | VGG-16 | 45.54 | 51.43 | 50.24 | 59.97 | 13.89 | 39.51 | 50.92 | - | 44.50 |
| PSPNet [40] | ResNet-50 | 50.73 | 64.68 | 54.32 | 63.41 | 26.55 | 40.68 | 47.45 | 65.43 | 49.69 |
| DeepLabv3+ [13] | ResNet-50 | 52.23 | 63.37 | 55.10 | 63.48 | 28.16 | 36.16 | 47.27 | 65.17 | 49.40 |
| OCRNet [75] | HRNetV2-W48 | 56.32 | 62.68 | 56.89 | 70.55 | 37.95 | 39.61 | 59.05 | 70.06 | 54.72 |
| SegFormer [76] | Mit-B0 | 54.05 | 63.91 | 53.78 | 68.33 | 30.46 | 41.00 | 52.36 | 67.55 | 51.98 |
| Segmenter [39] | Vit-B | 53.42 | 63.72 | 54.75 | 70.60 | 28.63 | 43.68 | 58.60 | 68.60 | 53.34 |
| MaskFormer [45] | ResNet-50 | 53.55 | 56.71 | 52.32 | 67.45 | 22.85 | 36.98 | 61.05 | 65.49 | 50.13 |
| Mask2Former [46] | ResNet-50 | 55.90 | 59.15 | 56.41 | 69.81 | 35.37 | 40.03 | 59.73 | 69.23 | 53.77 |
| CMTFNet [28] | ResNet-101 | 54.30 | 64.87 | 56.91 | 71.02 | 30.86 | 42.28 | 52.05 | 68.53 | 53.19 |
| RSSFormer [26] | RSS-B | 52.38 | 60.71 | 55.21 | 76.29 | 18.73 | 45.39 | 58.33 | - | 52.43 |
| MP-Former [77] | ResNet-50 | 55.49 | 63.09 | 55.52 | 70.56 | 33.76 | 37.10 | 61.00 | 69.34 | 53.79 |
| DenseCLIP [37] | ResNet-50 | 53.43 | 62.77 | 53.22 | 62.44 | 28.91 | 38.28 | 47.99 | 65.46 | 49.58 |
| SegCLIP | ResNet-50 | 53.91 | 66.22 | 58.48 | 70.24 | **41.27** | 42.14 | **61.65** | 71.44 | 56.27 |
| SegCLIP | Swin-B | **56.71** | **68.00** | **60.32** | **73.57** | 38.95 | **46.95** | 61.32 | **72.76** | **57.98** |

As the number of groups increases further, model performance sharply declines. Therefore, we opt for the optimal number of one GT text prompt group in our model design.

*5) Parameter Scales and Model Complexity:* Table IV shows a complexity comparison of the LoveDA dataset. For input data with a 1024 × 1024 pixel resolution, we measure the number of parameters and efficiency in millions (M) and floating point operations per second (FLOPs). The typical semantic segmentation model consists of a backbone and a segmentation head. Our method introduces a text encoder branch and improves the segmentation head, necessitating a measurement of network efficiency. Although we use only the 12-layer text encoder provided by CLIP, the parameter count for this component surpasses the combined total of the backbone and segmentation head, a limitation of our method. Besides adding a text encoder, our method also introduces improvements to the segmentation head. Notably, the parameter count for the segmentation head increases by only 2.02 million, while the FLOPs increase by 16.43 giga (G), and the mF1 and mIoU scores improve by 2.21% and 2.5%, respectively. Consequently, the primary reason for the increase in both parameters and FLOPs is the complexity of the language model structure. The proposed approach enhances the expressive power and learning potential of the model with a corresponding increase in the demand for computational resources and storage.

### D. Comparison With State-of-the-Art Methods

We compare the proposed method with a range of state-of-the-art semantic segmentation methods, including per-pixel classification methods like FCN [11], PSPNet [40], DeepLabv3+ [13], OCRNet [75], Segmenter [39], SegFormer [76], DenseCLIP [37], CMTFNet [28], and RSSFormer [26]. Additionally, we consider mask-based classification methods like MaskFormer [45], Mask2Former [46], and MP-Former [77].

*1) Results on the LoveDA Dataset:* The experimental results of various methods on the LoveDA dataset are shown in Table V. Compared to other semantic segmentation methods, SegCLIP significantly improves both mF1 and mIoU metrics, with mIoU being the more important metric. Under the same backbone network, SegCLIP achieves improvements of 2.21% in mF1 and 2.5% in mIoU, respectively, compared to Mask2Former. Additionally, it improves by 2.1% in mF1 and 2.48% in mIoU compared to MP-Former, a recent mask classification method. When compared with remote sensing semantic segmentation models, such as RSSFormer and CMTFNet, it records improvements of 3.84% and 3.08% in mIoU, respectively. For hard samples prone to misclassification by the model, the constructed text descriptions effectively capture the target representation, and the GTG further clarifies the target foreground region. Consequently, our method achieves significant improvements in the barren and forest classes compared to mask-based classification models. Furthermore, although the performance in the background class slightly lags behind Mask2Former, this is acceptable given the substantial overall improvement. We also present experimental results using Swin-B as the backbone network, where mF1 and mIoU reach 72.76% and 57.98%, respectively, significantly outperforming other models.

In Fig. 8, we show a visualization of the LoveDA dataset, selecting four representative samples to illustrate the superiority of our method, with boxes highlighting key areas. In the first row, generally, trees or low vegetation on roads

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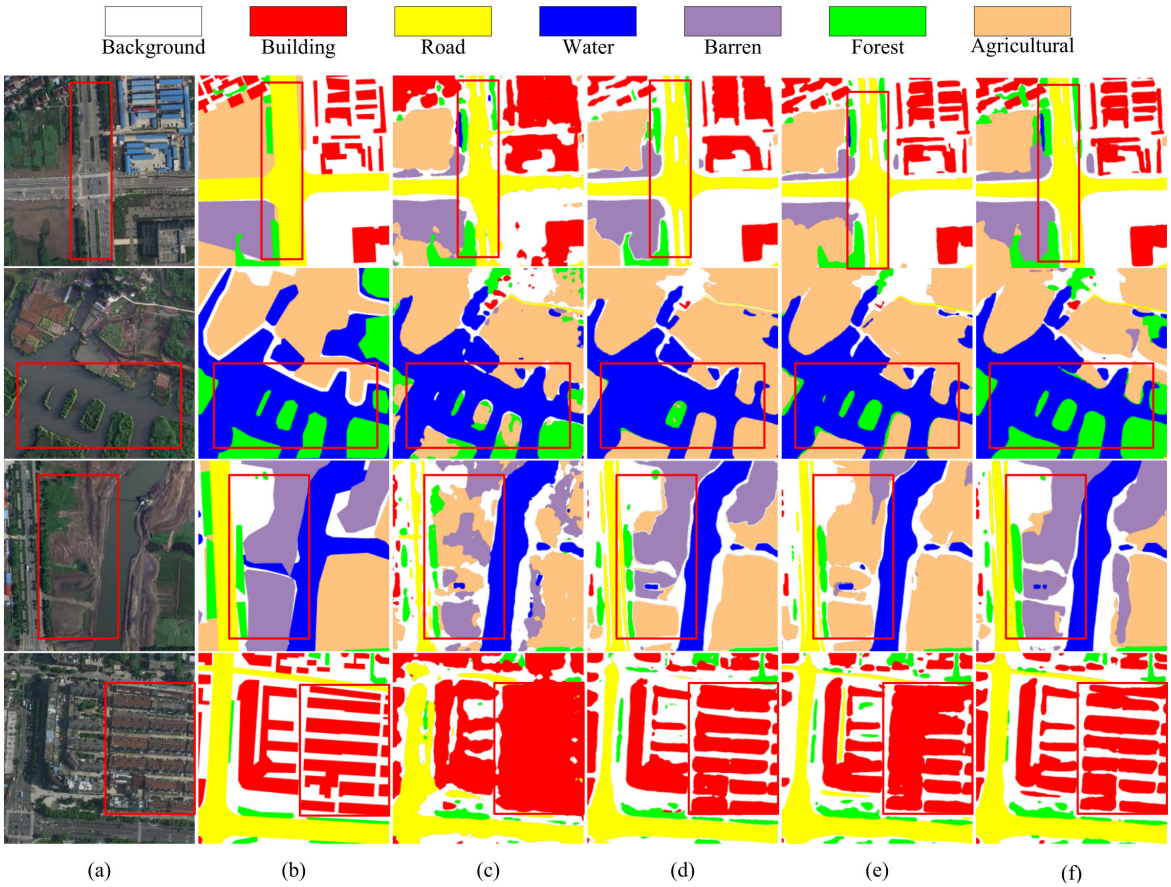



Fig. 8. Our method is based on the ResNet-50 backbone and is qualitatively compared with other methods on the LoveDA dataset. (a) RGB image. (b) GT. (c) MaskFormer. (d) Mask2Former. (e) MP-Former. (f) SegCLIP (ours).

are often classified as part of the road landscape; however, other methods mistakenly classify these as background. In contrast, our method provides relatively accurate predictions. This indicates that SegCLIP effectively integrates text information with visual features, enhancing the ability to discern intracategory differences in complex scenes. Additionally, the third row reveals class similarity between barren on the left and agriculture on the right of the water class, with only SegCLIP achieving accurate categorization. Moreover, other rows further demonstrate the ability of SegCLIP to accurately differentiate between classes. In summary, qualitative comparisons in complex scenarios reveal that SegCLIP delivers more accurate segmentation results than other methods, owing to the integration of multimodal text information with visual features.

*2) Results on the iSAID Dataset:* Compared to the LoveDA dataset, the primary challenge of iSAID as a large-scale, fine-grained semantic segmentation benchmark in remote sensing lies in addressing complex scale diversity and extreme foreground segmentation. As shown in Table VI, under the unified backbone network, we achieve a significant breakthrough in the helicopter (HC) class compared to other mask classification methods. Compared to Mask2Former, our method improves the IoU for the HC class from 29.01% to 32.17%. Compared to MP-Former, our method increases the IoU for the HC class by 9.7%. Given the significant visual similarity between helicopter (HC) and plane (PL), mask-based segmentation models struggle to distinguish these differences. Our approach effectively uses text information to explicitly discern differences between classes, thus enhancing the segmentation performance of hard samples. Our SegCLIP surpasses other mask classification methods in all classes except for baseball diamonds (BD), basketball court (BC), swimming pool (SP), and roundabout (RA). When the backbone is replaced with Swin-B, our method achieves state-of-the-art performance, registering 67.78% and 79.68% in mIoU and mF1, respectively.

In Fig. 9, we show a visualization of the iSAID dataset, selecting four representative samples to illustrate the superiority of our method, with boxes highlighting key areas. In the first row, helicopter and plane exhibit a high degree of visual similarity, making it difficult for other methods to distinguish between them, which leads to missed detections or misclassifications. By fusing multimodal information, our method effectively distinguishes feature differences between classes, primarily due to the consistent learning of text information cues and visual features, resulting in accurate segmentation. In the fourth row, the red box clearly shows a U-shaped road, which does not belong to the roundabout class and should be categorized as background. Other methods commonly misclassify it as a roundabout. Our visualization results are more accurate and capable of segmenting remote sensing images accurately in complex scenes. Additionally, the samples in the

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TABLE VI

QUANTITATIVE COMPARISONS WITH THE STATE-OF-THE-ART METHODS ON THE iSAID DATASET. THE METRIC FOR ALL CATEGORIES IS IoU

| Method | Backbone | Per-class IoU(%) | | | | | | | | | | | | | | | mF1 | mIoU |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | SH | ST | BD | TC | BC | GTF | BR | LV | SV | HC | SP | RA | SBF | PL | HA | (%) | (%) |
| FCN [11] | ResNet-101 | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | | 44.6 |
| PSPNet [40] | ResNet-101 | 59.71 | 66.27 | 62.98 | 82.05 | 52.11 | 53.45 | 31.67 | 56.61 | 42.74 | 29.65 | 39.89 | 67.09 | 72.89 | 75.41 | 53.82 | - | 59.07 |
| DeepLabv3+ [13] | ResNet-101 | 62.25 | 67.93 | 75.84 | 85.21 | 60.60 | 51.14 | 42.88 | 58.95 | 42.85 | 32.27 | 46.08 | 61.32 | 73.33 | 76.73 | 53.07 | - | 61.82 |
| OCRNet [75] | HRNetV2-W48 | 63.72 | 70.02 | 73.76 | **87.69** | 52.18 | 59.76 | 40.84 | 59.26 | 42.99 | 21.91 | 46.20 | 75.04 | 74.28 | 78.88 | 53.67 | - | 62.45 |
| SegFormer [76] | Mit-B0 | 68.10 | 71.93 | 68.66 | 86.10 | 54.13 | 55.63 | 35.94 | 61.82 | 46.46 | 30.89 | 47.34 | 52.78 | 72.70 | 82.07 | 55.13 | 74.92 | 61.80 |
| Segmenter [39] | Vit-B | 61.19 | 67.28 | 75.41 | 84.83 | 59.95 | 61.79 | 36.30 | 56.84 | 35.75 | **38.74** | 43.98 | 67.26 | 76.47 | 76.56 | 49.52 | 75.02 | 61.92 |
| MaskFormer [45] | ResNet-50 | 65.62 | 65.63 | 64.44 | 84.74 | 58.21 | 48.35 | 37.39 | 63.27 | 50.27 | 29.01 | 43.54 | 73.00 | 73.41 | 83.82 | 55.80 | 75.20 | 62.22 |
| Mask2Former [46] | ResNet-50 | 67.37 | 65.77 | 65.96 | 84.31 | 46.01 | 56.74 | 43.28 | 64.73 | 52.80 | 19.43 | 48.77 | 76.24 | 68.04 | 83.98 | 56.14 | 75.15 | 62.40 |
| CMTFNet [28] | ResNet-101 | 65.02 | 71.67 | 77.01 | 87.62 | 61.37 | 62.71 | 40.19 | 59.98 | 43.68 | 16.94 | 47.35 | 72.74 | 76.71 | 80.48 | 55.49 | - | 63.63 |
| RSSFormer [26] | RSS-B | 65.91 | 73.55 | **79.68** | 86.92 | **64.41** | 61.13 | 44.76 | 60.67 | 41.34 | 35.08 | 50.22 | **78.28** | 78.57 | 73.64 | 55.60 | - | 65.55 |
| MP-Former [77] | ResNet-50 | 66.66 | 67.87 | 67.36 | 85.34 | 60.57 | 58.56 | 44.05 | 65.11 | 53.46 | 22.47 | 48.46 | 75.79 | 62.42 | 84.88 | 57.01 | 76.33 | 63.67 |
| DenseCLIP [37] | ResNet-50 | 64.07 | 55.06 | 61.23 | 85.63 | 61.53 | 50.71 | 36.77 | 53.03 | 47.88 | 22.96 | **62.16** | 52.04 | 76.05 | 73.23 | 46.41 | 72.81 | 59.23 |
| SegCLIP | ResNet-50 | 70.45 | 68.06 | 66.90 | 86.93 | 58.55 | 59.23 | 44.82 | 65.22 | 53.50 | 32.17 | 46.76 | 75.58 | 74.29 | 85.43 | 59.17 | 77.84 | 65.39 |
| SegCLIP | Swin-B | **73.48** | **74.24** | 68.98 | 87.66 | 62.49 | **65.49** | **45.55** | **68.20** | **54.27** | 38.61 | 44.99 | 74.88 | **79.11** | **86.60** | **62.83** | **79.68** | **67.78** |

Note: SH: ship, ST: storage tank, BD: baseball diamond, TC: tennis court, BC: basketball court, GTF: ground track field, BR: bridge, LV: large vehicle, SV: small vehicle, HC: helicopter, SP: swimming pool, RA: roundabout, SBF: soccer ball field, PL: plane, HA: harbor.

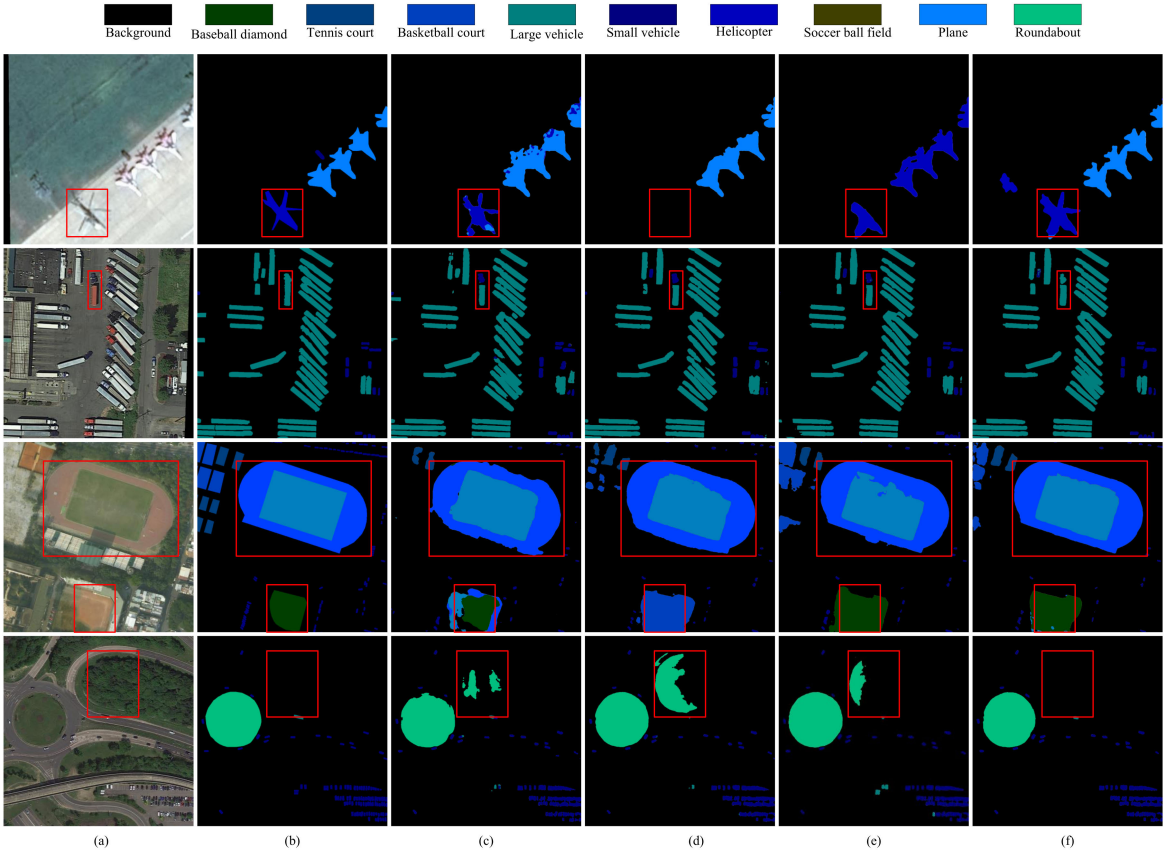



Fig. 9. Our method is based on the ResNet-50 backbone and is qualitatively compared with other methods on the iSAID dataset. (a) RGB image. (b) GT. (c) MaskFormer. (d) Mask2Former. (e) MP-Former. (f) SegCLIP (ours).

other rows further validate the accuracy of our method in the semantic segmentation of large-scale remote sensing images, fully demonstrating the importance of multimodal information in remote sensing semantic segmentation.

*3) Results on the UAVid Dataset:* Compared to the LoveDA and iSAID datasets, the UAVid dataset emphasizes dynamic semantic segmentation scenarios, particularly focusing on the recognition of moving targets and the preservation of temporal consistency. The results, as shown in Table VII, indicate that our method achieves higher performance compared to the mask classification method when using ResNet-50 as the backbone network. Compared to Mask2Former, all metrics are significantly improved, with mF1 and mIoU increasing by 0.59% and 0.78%, respectively. Compared to the recent

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TABLE VII
QUANTITATIVE COMPARISONS WITH THE STATE-OF-THE-ART METHODS ON THE UAVID DATASET. THE METRIC FOR ALL CATEGORIES IS IOU

| Method | Backbone | Per-class IoU(%) | | | | | | | | mF1 | mIoU |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Background | Building | Road | Tree | Low Vegetation | Moving Car | Static Car | Human | (%) | (%) |
| FCN [11] | ResNet-101 | 61.36 | 90.27 | 74.02 | 77.37 | 65.66 | 72.47 | 68.17 | 36.07 | 80.08 | 68.17 |
| PSPNet [40] | ResNet-50 | 67.04 | 92.27 | 80.07 | 78.52 | 68.53 | 73.27 | 71.62 | 42.35 | 82.75 | 71.71 |
| DeepLabv3+ [13] | ResNet-50 | 66.97 | 92.43 | 79.04 | 78.39 | 67.61 | 74.49 | 71.39 | 40.30 | 82.41 | 71.33 |
| OCRNet [75] | HRNetV2-W48 | 66.75 | 92.52 | 79.71 | 78.83 | 69.44 | 75.57 | 71.39 | 40.66 | 82.78 | 71.86 |
| SegFormer [76] | Mit-B0 | 66.35 | 91.86 | 79.28 | 78.47 | 68.79 | 75.56 | 69.18 | 41.99 | 82.55 | 71.44 |
| Segmenter [39] | Vit-B | 67.98 | 92.59 | 80.62 | 78.18 | 69.42 | 72.98 | 67.32 | 32.48 | 81.24 | 70.20 |
| MaskFormer [45] | ResNet-50 | 59.87 | 89.82 | 74.04 | 76.16 | 66.33 | 73.73 | 67.37 | 41.04 | 80.55 | 68.54 |
| Mask2Former [46] | ResNet-50 | 68.00 | 92.54 | 80.64 | 78.77 | 68.68 | 77.37 | 72.57 | 44.63 | 83.62 | 72.90 |
| CMTFNet [28] | ResNet-101 | 67.22 | 92.12 | 80.84 | 78.80 | 70.38 | 74.43 | 70.82 | 47.11 | 83.59 | 72.71 |
| RSSFormer [26] | RSS-B | 67.23 | 92.47 | 81.48 | 78.00 | 70.56 | 70.16 | 67.13 | 38.49 | 81.88 | 70.69 |
| MP-Former [77] | ResNet-50 | 68.36 | 92.45 | 80.41 | 79.70 | 70.79 | 75.64 | 71.54 | 45.50 | 83.76 | 73.04 |
| DenseCLIP [37] | ResNet-50 | 65.68 | 91.85 | 78.08 | 78.20 | 68.35 | 75.02 | 71.26 | 43.89 | 82.71 | 71.54 |
| SegCLIP | ResNet-50 | 68.44 | 92.92 | 81.14 | 79.21 | 70.41 | 77.99 | 72.85 | 46.52 | 84.21 | 73.68 |
| SegCLIP | Swin-B | **70.58** | **93.84** | **82.22** | **80.57** | **72.72** | **79.65** | **75.88** | **50.40** | **85.66** | **75.73** |

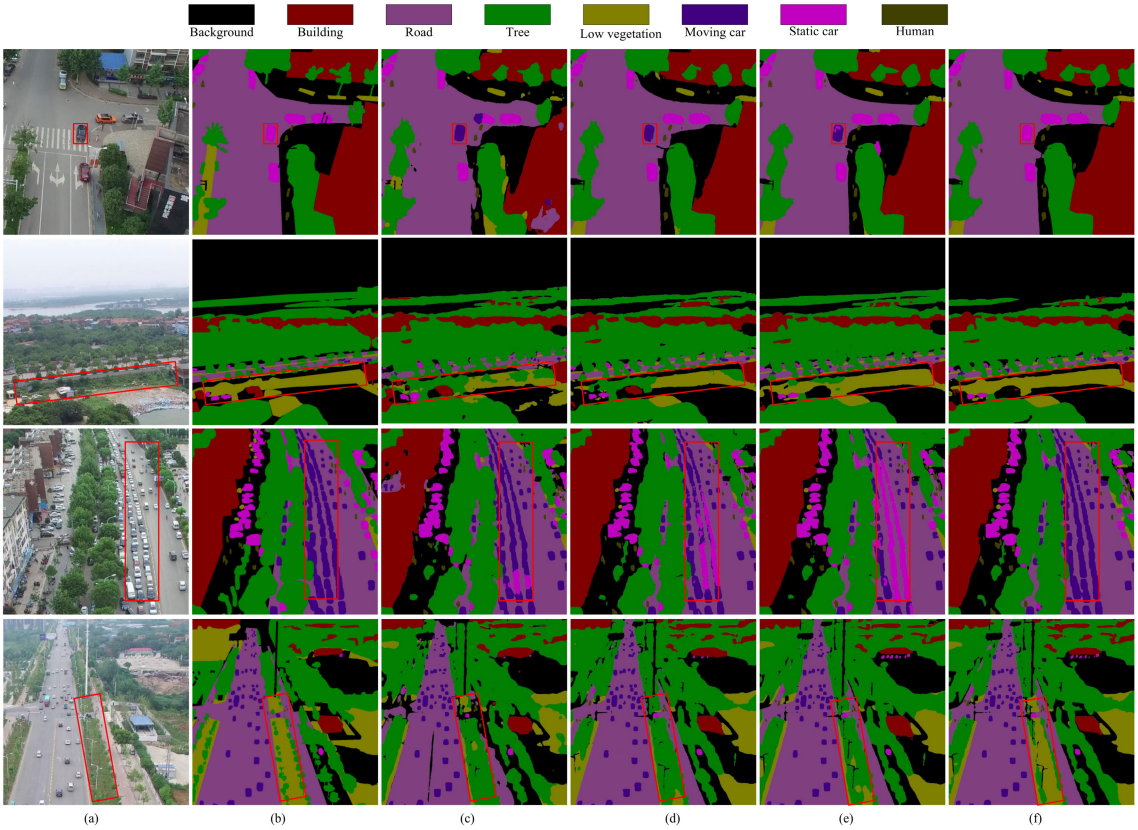



Fig. 10. Our method is based on the ResNet-50 backbone and is qualitatively compared with other methods on the UAVid dataset. (a) RGB image. (b) GT. (c) MaskFormer. (d) Mask2Former. (e) MP-Former. (f) SegCLIP (ours).

MP-Former, mF1 and mIoU increase by 0.45% and 0.64%, respectively. Notably, the performance of our method is particularly significant in the human class, consistent with its performance on other datasets, thereby demonstrating its effectiveness in improving performance on hard samples. When using Swin-B as the backbone network, our method significantly outperforms other models across all metrics, with mF1 and mIoU reaching 85.66% and 75.73%, respectively.

In Fig. 10, we show a visualization of the UAVid dataset, selecting four representative samples to illustrate the superiority of our method, with boxes highlighting key areas. In the first and third rows, regardless of whether the cars are moving or stationary, they tend to have similar exterior designs and colors. These common factors make it challenging to determine their dynamic state (moving or static) from visual observation alone. Differentiating by human eye observation alone is difficult due to the high visual similarity that masks the differences in motion states. Among all methods, only ours accurately visualizes the results. By introducing linguistic descriptive information, such as “a photo of a moving car”

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TABLE VIII
EFFICIENCY EVALUATION ON THE LOVEDA DATASET

| Method | Backbone | Inference time(ms) |
|---|---|---|
| FCN [11] | ResNet-101 | 66.67 |
| PSPNet [40] | ResNet-50 | 50.58 |
| DeepLabv3+ [13] | ResNet-50 | 47.86 |
| OCRNe [75] | HRNetV2-W48 | 72.76 |
| SegFormer [76] | Mit-B0 | 33.71 |
| Segmenter [39] | Vit-B | 134.46 |
| MaskFormer [45] | ResNet-50 | 38.41 |
| Mask2Former [46] | ResNet-50 | 61.03 |
| CMTFNet [28] | ResNet-101 | 25.46 |
| RSSFormer [26] | RSS-B | 82.33 |
| MP-Former [77] | ResNet-50 | 61.67 |
| DenseCLIP [37] | ResNet-50 | 74.19 |
| SegCLIP | ResNet-50 | 82.81 |
| SegCLIP | Swin-B | 115.53 |

or "a photo of a static car," our model effectively relies on text information to discern the difference between the two. Furthermore, the comparison results in the second and fourth rows demonstrate the superior performance of our proposed SegCLIP model in executing the semantic segmentation task. SegCLIP not only accurately recognizes the dynamics of cars but also improves the accuracy of object segmentation within the scene, thereby verifying the effectiveness of our method in understanding and distinguishing the dynamic characteristics of objects in complex scenes.

### E. Efficiency Analysis

Table VIII details the inference times for the various semantic segmentation models covered in this study on the LoveDA dataset. Even though our SegCLIP model doubles the number of parameters due to the addition of text prompt branches, its inference time is only 21.78 ms longer than that of Mask2Former. Despite the additional computational burden, SegCLIP still demonstrates a significant advantage in inference efficiency, especially compared to the Segmenter model based on per-pixel classification. Meanwhile, the gap between SegCLIP and RSSFormer is relatively small.

In summary, although the introduction of CLIP inevitably affects inference speed, this impact remains within acceptable limits. Therefore, the method of combining linguistic description with visual feature extraction is not only feasible in remote sensing semantic segmentation tasks but also holds potential for further exploration and application.

## V. CONCLUSION

In this article, we propose a multimodal semantic segmentation framework called SegCLIP. CLIP is introduced into the HRRS semantic segmentation task, enhancing segmentation ability through the integration of text information and visual features of remote sensing images. The proposed TMP uses category-related text hints to significantly narrow the query scope between categories. To further use text information efficiently, we thoroughly explore its potential, and the proposed GTG employs GT text description guidance to optimize the query process. Additionally, a simple CFF module is designed to unify the contextual relationship between visual features and text information. Therefore, the proposed SegCLIP is a multimodal mask classification method for text prompt queries. We conduct extensive experiments on three HRRS datasets (LoveDA, iSAID, and UAVid) to verify the effectiveness of the proposed method.

Since large language models bring too many parameters, in the future, we will explore lightweight frameworks for semantic segmentation of multimodal remote sensing images.

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